LAB EXAM 1st batch

Reg. No. | | | | | | | | | | |

[13CEL69/12CEL69]

SIXTH SEMESTER B.Tech./B.E EXAMINATION, JUNE – 2016

DESIGN AND DRAWING OF STRUCTURES

Time: 3 Hours

Maximum Marks: 70

Instructions:

1. Assume suitable non trivial data wherever necessary.
2. Use of IS456:2000 is allowed. Use of SP-16 is allowed.
3. Answer any **ONE** full Question each from both part A and part B.

PART - A

1. Design a RC cantilever retaining wall to retain earth level with the top of the wall to a height of 4m high above ground level. The unit weight of earth is 18KN/m^3 and its angle of repose is 30°. The embankment is horizontal at its top. The SBC of soil can be considered as 100KN/m^2 and coefficient of friction between soil and concrete is 0.5. Use M-20 mix and Fe-415 bars. Draw Plan of base slab. LS & CS retaining wall. 30 20

2. Design a combined rectangular footing for two columns A and B, carrying loads of 1000kN and 1000kN, spaced at 4.5m apart centre to centre. The projection of the footing beyond the centre of each column is 1.25m. The columns are 400mm × 400mm each. The safe bearing capacity of soil is 200KN/m^2. Use M-20 concrete and Fe415 steel. Draw plan of footing, LS & CS showing all details. 30 20

PART – B

3. Draw a plate girder for the following data: 20

Span =30m, simply supported

Web plate = 200mm × 12mm

Flange plate = 700mm × 50mm

End bearing stiffeners = 220mm × 20mm

Intermediate stiffeners = 200mm × 20mm

Spacing of intermediate stiffners = 2000mm

All connections are welded.

Adopt 5mm shop fillet weld.

Draw plan, elevation and cross-section choosing all details.

4. Draw the plan and elevation with cross section details showing all details of reinforcement for the column footing with following data: 20

Footing size = 3.4m × 3.4m

Column size = 500mm × 500mm

Column Reinforcement = 8 bars of 20mm

Footing reinforcement = 24 bars of 12mm uniformly spaced in both directions

Depth of footing varies from 760 mm at the face of column to 200mm at free end.

Column strirrups = 8mm bars @ 150c/c